

4	(a) kinetic energy = $\frac{1}{2}mv^2$ $= \frac{1}{2} \times 0.040 \times (2.8)^2 = 0.157 \text{ J or } 0.16 \text{ J}$	C1	A1	[2]
	(b) (i) $k = F/x$ or $F = kx$ $X_B = 14/800$ $= 0.0175 \text{ m}$	C1	A1	[2]
	(ii) area under graph = elastic potential energy stored or $\frac{1}{2}kx^2$ or $\frac{1}{2}Fx$ (energy stored =) 0.1225 J less than KE (of 0.16 J)	C1	A1	[2]